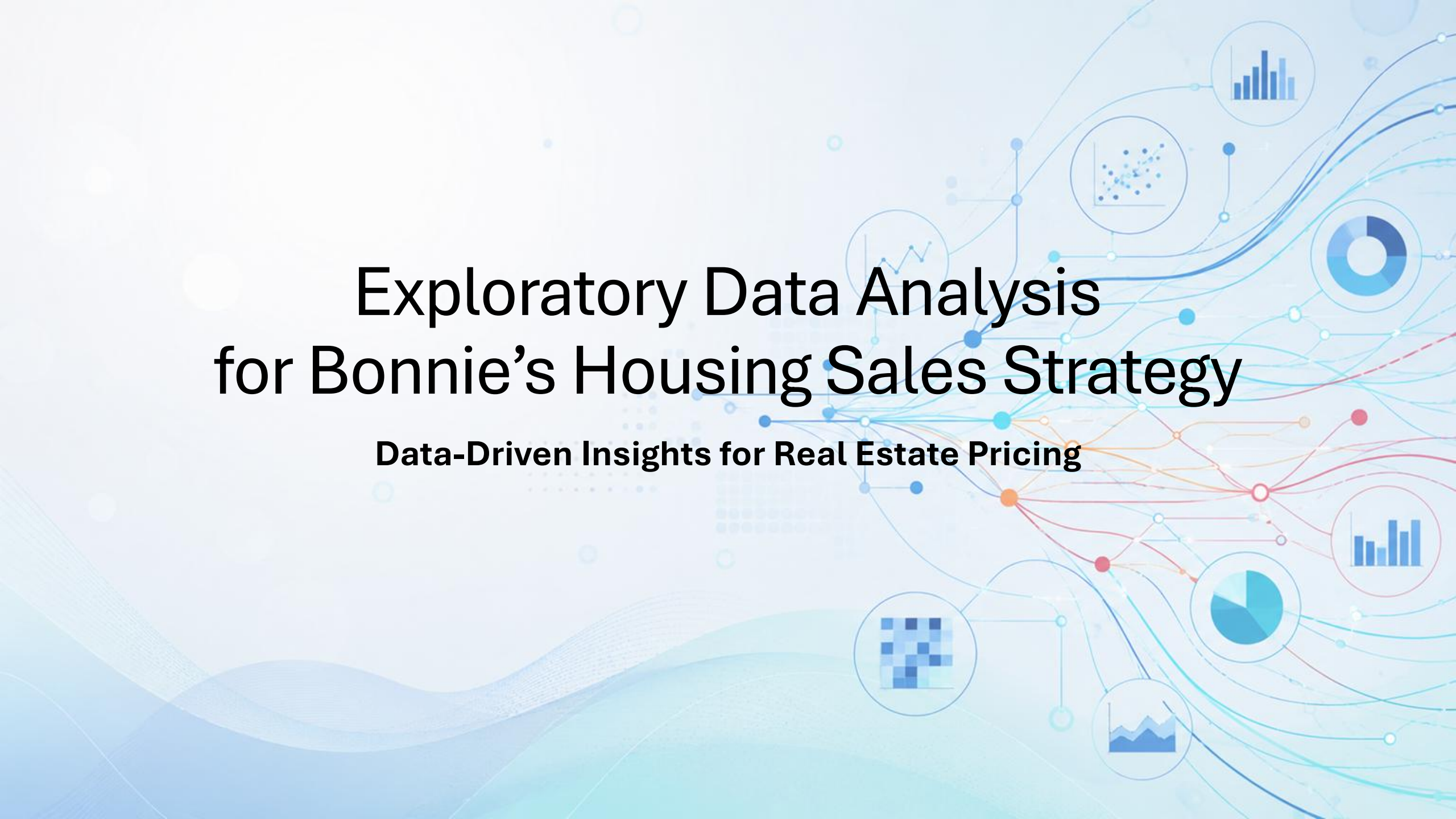


# Exploratory Data Analysis for Bonnie's Housing Sales Strategy

**Data-Driven Insights for Real Estate Pricing**



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# Business Problem



# Business Problem & Goal

## Goal of the Analysis

- Identify key price factors
- Find the best selling period
- Analyze regional market segments
- Support Bonnie's selling strategy



# Dataset Overview

- 21,000+ house sales
- Regional & seasonal data
- Multiple price-related features
- Some missing values (NaN)

|   | Feature     | Meaning         |
|---|-------------|-----------------|
| 0 | price       | selling price   |
| 1 | sqft_living | living space    |
| 2 | grade       | overall quality |
| 3 | zipcode     | location        |
| 4 | view        | quality of view |
| 5 | waterfront  | water access    |
| 6 | condition   | house condition |
| 7 | month       | month of sale   |

# Analysis Approach

# Analysis Approach

- Data exploration
- Hypothesis creation
- Regional analysis
- Seasonal analysis
- Correlation analysis
- Data cleaning



# Analysis Approach

- Larger houses → higher prices
- Region influences price
- Spring may increase prices
- Grade influences price

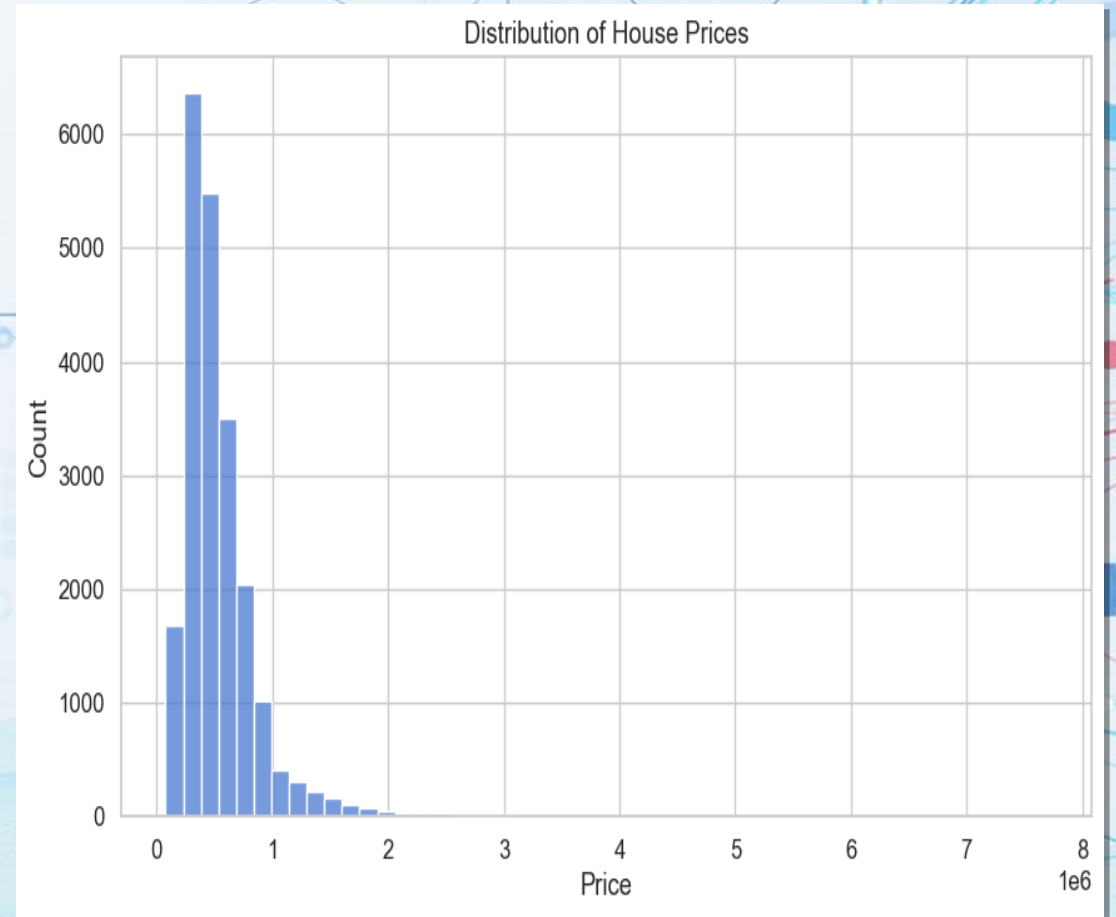


# Market Overview



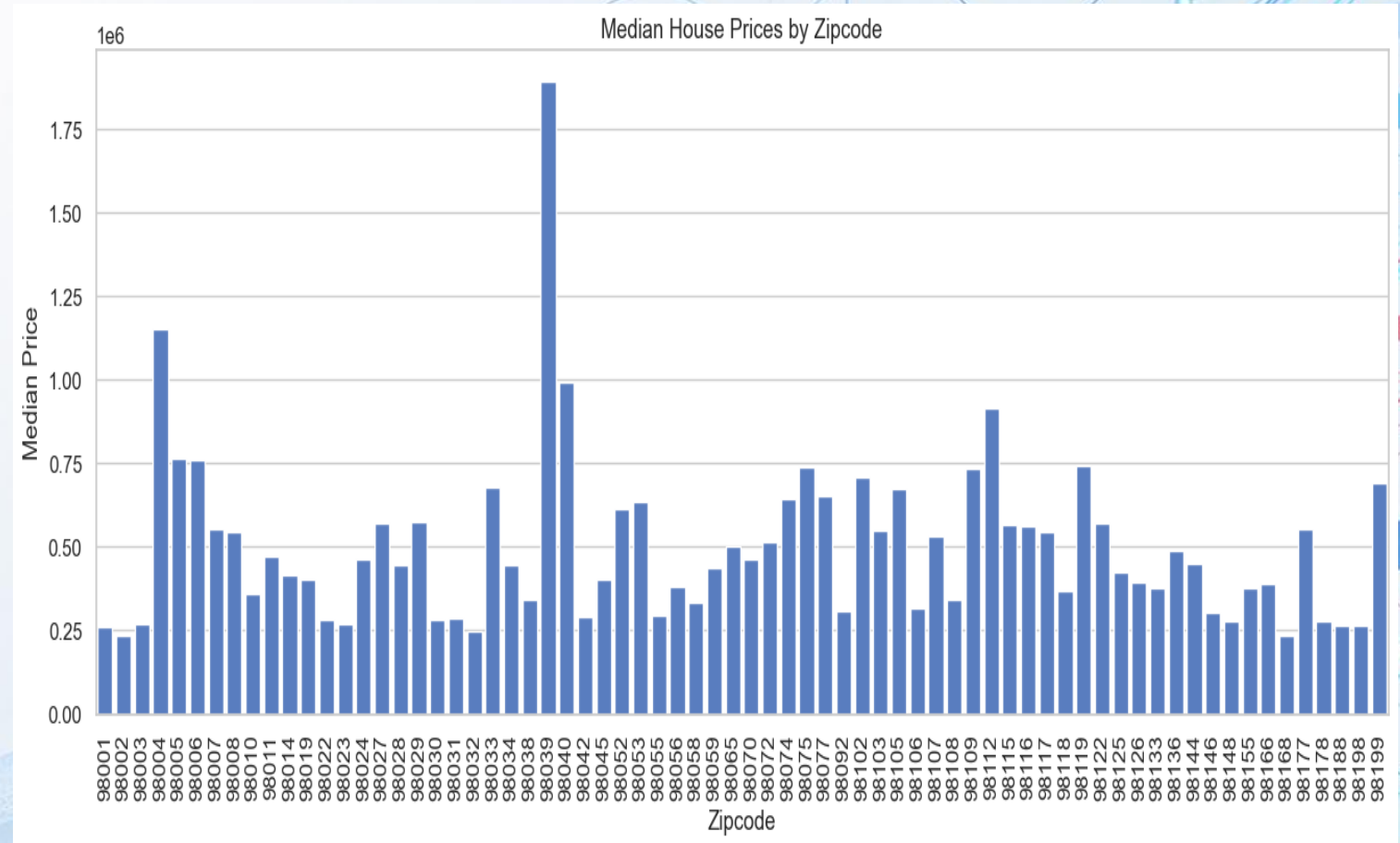
# Market Overview

- Strong right-skewed distribution
- Small number of luxury properties
- Median values preferred over averages



# Regional Market Analysis

- Strong differences between zipcodes
- Multiple market segments identified
- Location strongly influences prices

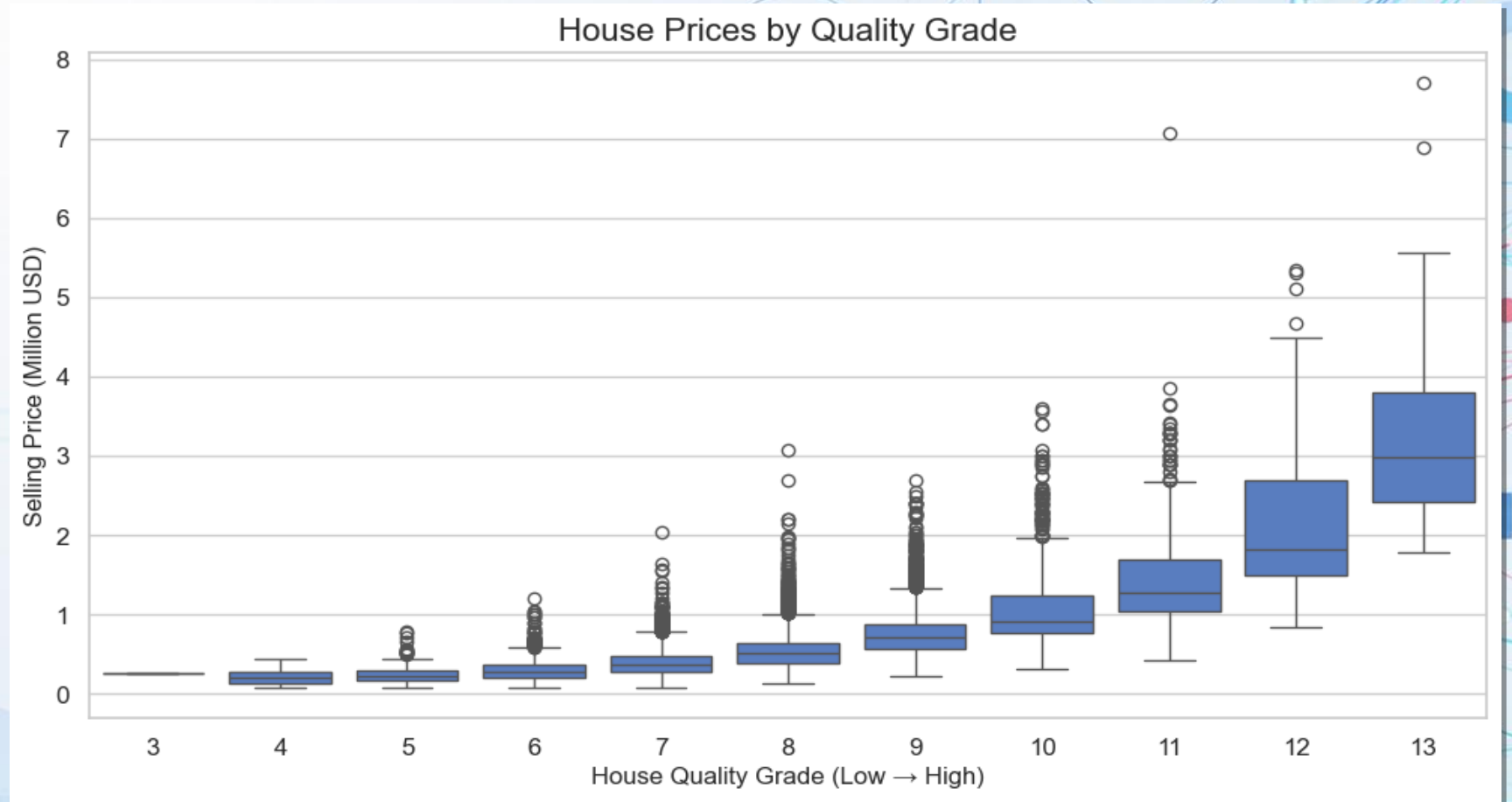


# Key Findings & Recommendations



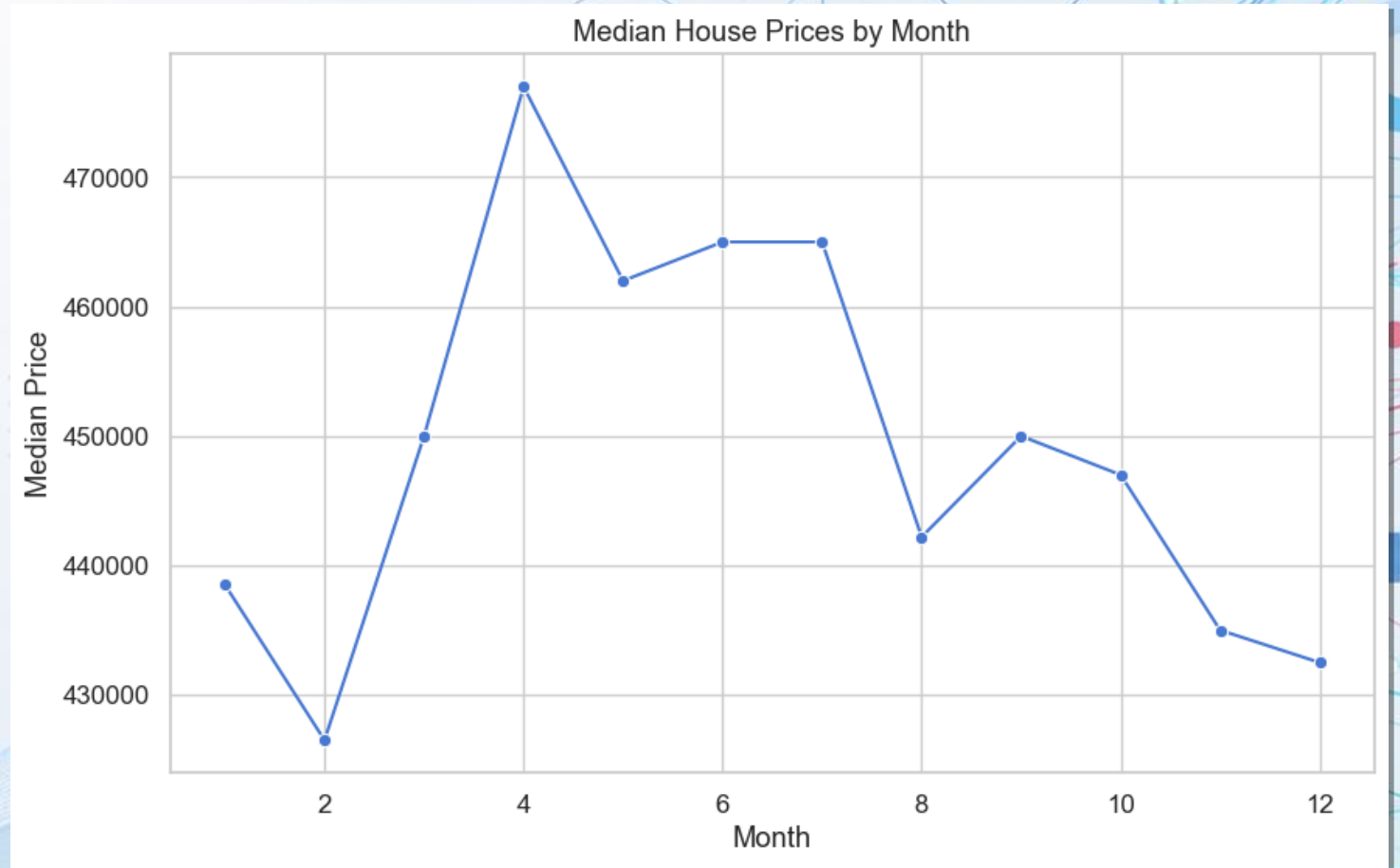
# Key Price Factors

- Higher grade → higher prices
- Strong relationship between quality and price
- Luxury houses create larger price ranges



# Best Selling Season

- Prices rise during spring
- Peak around April and early summer
- Prices decrease toward winter



# Final Recommendations

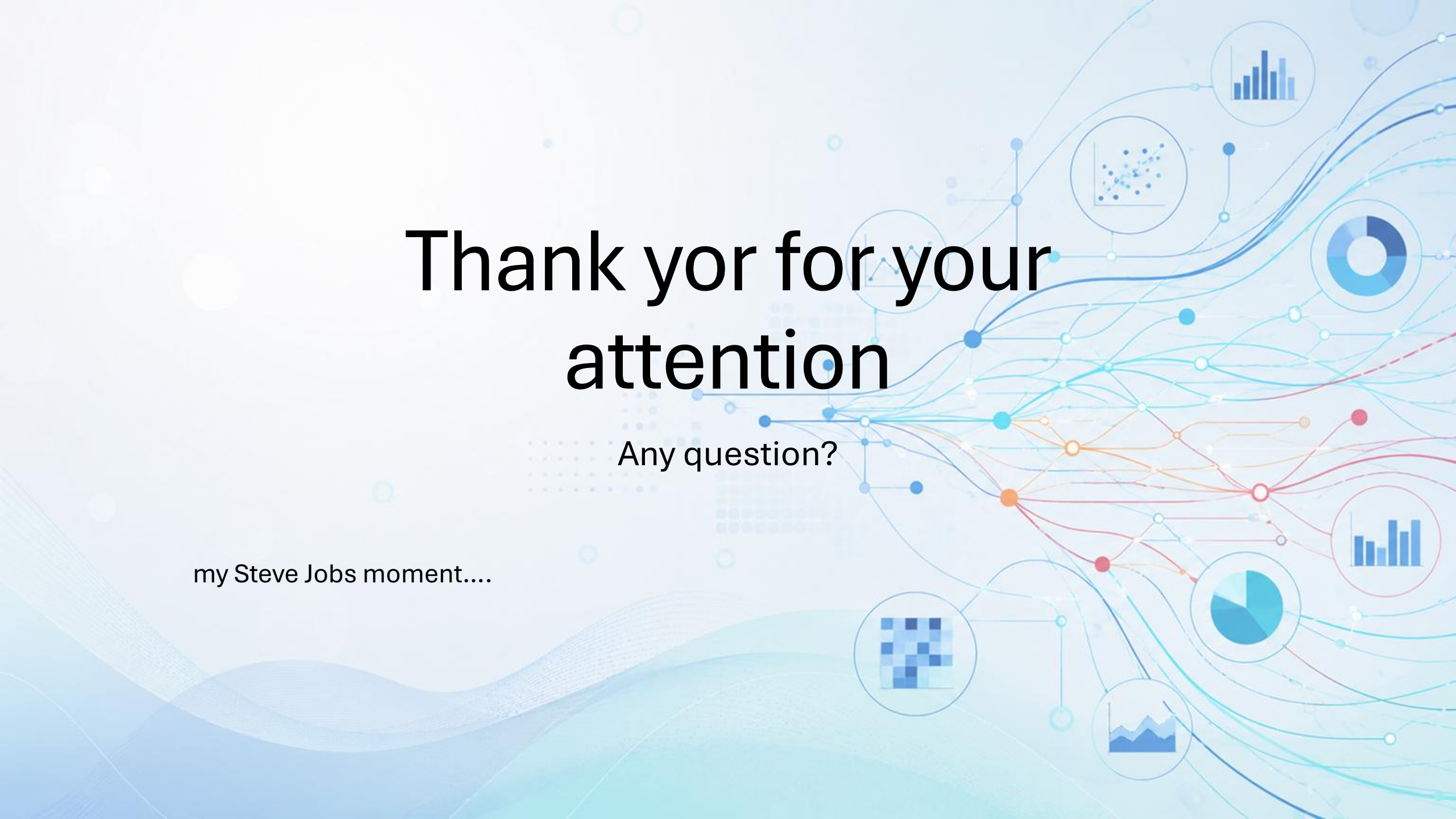
- Prefer spring or early summer sales
- Use regional comparisons
- Focus on property presentation
- Use median values for market evaluation



# Conclusion

- Location strongly influences prices
- Grade and living space are key factors
- Seasonal timing impacts selling prices
- Data-driven analysis supports better decisions

*Data-driven decisions can improve selling strategies  
and market positioning.*



# Thank you for your attention

Any question?

my Steve Jobs moment....

# BONUS SLIDE

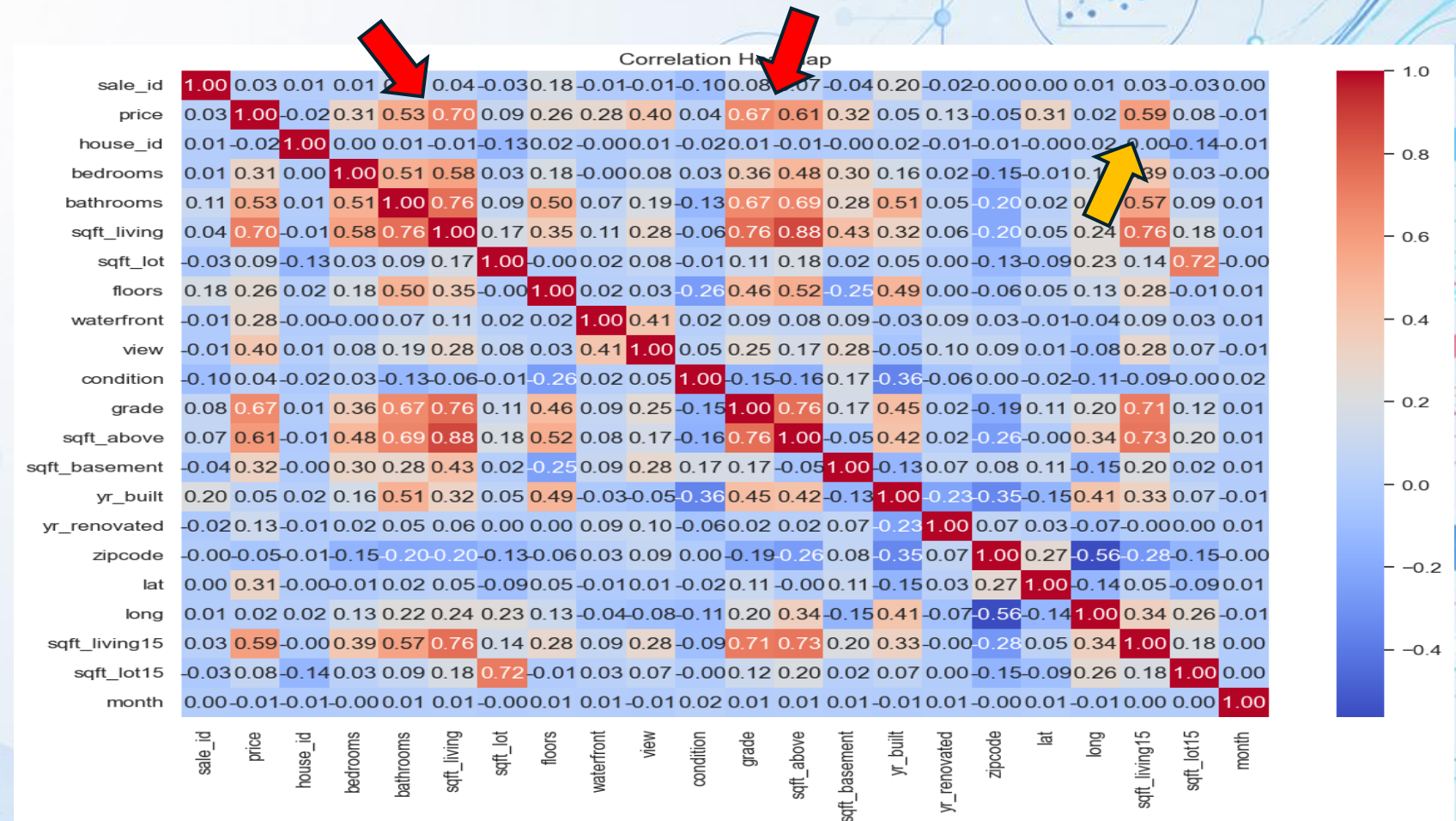
One more thing...



# Correlation Heatmap

## Key Insights:

- Strongest factors:
  - living space
  - grade
- Correlation  $\neq$  causation



# End

Enjoy the day

